

## Sleep organization pattern as a prognostic marker at the subacute stage of post-traumatic coma

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### Abstract

**Objectives:** The aim of this study is to identify the predictive indexes for post-traumatic coma prognosis, which is important to better direct acute and subacute treatments and rehabilitation efforts. The pattern of sleep organization is a potential prognostic marker, but its role has not been established yet in the context of modern critical care. In the present study, we used a new protocol to evaluate the prognostic value of the different levels of sleep–wake organization recorded at the subacute stage of post-traumatic coma.

**Methods:** Twenty-four head-injured comatose patients were monitored with 24 h polysomnographic recordings. The predictivity of the different levels of sleep–wake organization on polysomnography was compared with other possible prognostic indexes (i.e. neuroradiological findings, age and Glasgow Coma Scores (GCS)). Main outcome measures were survival and the degree of disability after recovery from coma.

**Results:** The presence of organized sleep patterns, but not GCS, was highly predictive of better outcome (odds ratio = 10.78,  $P = 0.01$ ), even after correction for potentially confounding variables with multivariate analysis.

**Conclusions:** Our study demonstrates that the sleep–wake organization pattern based on 24 h polysomnographic recordings at the subacute stages of post-traumatic coma is a reliable prognostic marker, both for survival and for functional recovery. © 2002 Elsevier Science Ireland Ltd. All rights reserved.

**Keywords:** Post-traumatic coma; Sleep; Prognosis; Polygraphy; Electroencephalogram

### 1. Introduction

Modern intensive care permitted more comatose patients to survive after severe head injury, even though often with severe disabilities. It is very important to identify these patients so as to better direct acute and subacute treatments and rehabilitation efforts, thus improving cost/benefit ratios. Traditionally, Glasgow Coma Scale (GCS) has been used as a reliable predictor of clinical outcome in head-injured comatose patients (Jennett et al., 1976; Jennett and Teasdale, 1981). However, this method of evaluation finds severe limitations in the fact that it does not take into account the responses of brainstem reflexes. In fact, other scales, such as the Innsbruck Coma Scale, which includes the evaluation of some brainstem reflexes, demonstrated

higher sensitivity for non-survival prediction (Benzer et al., 1991). Anyway, it can be difficult to assign coma scale scores in the comatose patient in case of multiple severe injuries or of artificial ventilation. In addition, contrary to what is observed at the very acute stage of coma, there remains a lack of reliable methods to assess prognosis at the subacute and recovery stages of traumatic coma.

A possible prognostic marker that has raised interest in the past few years is the pattern of sleep organization. Electroencephalogram (EEG) patterns similar to those of sleep have been considered as favorable prognostic markers (Chatrian et al., 1963; Bergamasco et al., 1968a,b; Bricolo and Turella, 1973; Bricolo et al., 1979; Rumpl et al., 1983; Evans and Bartlett, 1995). On the contrary, the presence of exclusively rudimentary forms of oscillation has been described in association with poor outcome (Bergamasco et al., 1968a,b; Fischgold and Mathis, 1959). Sleep pattern has been shown to continue to improve during rehabilita-

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tion, and REM sleep percentage increases with recovery of cognitive functions (Harada et al., 1976; Ron et al., 1980). In a series of 10 patients with post-traumatic apallic syndrome, the absence of a definite sleep pattern was observed exclusively in the only patient who never aroused from a persistent vegetative state and who eventually died before the sixth month of follow-up (Giubilei et al., 1995). Recently, the problem of sleep patterns in head-injured patients was readdressed by Evans and Bartlett (1995), with special attention to the presence or absence of arousal-related phasic activity involving EEG.

Despite these relevant contributions, the study of hypnic organization of comatose patients is not exhaustive, and polysomnography (PSG) is not usually performed as part of the routine in head-injured patients in most intensive care units (ICU). In fact, scoring criteria of past studies are no longer acceptable and previous imaging limitations did not allow precise evaluation of findings according to damage localization. On the other hand, more recent studies do not consider the cyclicity of sleep and present only an instant picture of EEG, assuming that the comatose patient shows no circadian oscillation of its encephalic electrical activity. In particular, except for the apallic syndrome, there are no studies using 24 h recordings, which is the only method that allows the assessment of the possible time dispersion of sleep periods, due to the brain damage itself and/or the lack of temporal indicators that patients experience in ICUs. In addition, many studies were mainly observational and/or lacked solid statistical analysis, especially with reference to the role of possible confounding factors.

Based on these considerations, we decided to reconsider the topic using a protocol of 24 h PSG monitoring, aimed at better defining the prognostic value of the different patterns of sleep–wake organization recorded in patients at the subacute stages of coma after severe head injury.

## 2. Materials and methods

The study was conducted on a series of consecutive patients admitted to the intensive care unit of the ‘A. Gemelli’ University Hospital (Rome) for severe head injury. Within less than 24 h after admission, all the patients were evaluated with neurological examination, Glasgow Coma Scale (GCS) (Jennett et al., 1976; Jennett and Teasdale, 1981), brain computer tomographic (CT) scan and EEG. Patients with GCS >9 at admission, with seizures or with clinical/EEG signs of brain death were excluded from the study. During the acute period, patients were submitted to propofol sedation (0.5–2 mg/kg/h) and to analgesia with fentanyl whenever required. Barbiturates and benzodiazepines were not used. After initial selection, the study group included 24 patients (19 males and 5 females, mean age  $41 \pm 17.1$ , range 16–74 years). At the subacute stage, at least 24 h after sedative and neuroprotective drugs withdrawal, a 24 h polysomnographic recording was performed on

magnetic cassette (Oxford Medilog 9200), with an interval ranging from 7–14 days following injury. GCS was calculated again at this stage, immediately before PSG recording. Montage included 4 or 5 EEG channels, with electrodes placed according to the 10-20 International System (C4-A1, C3-A2, P4-O2, P3-O1), two electrooculogram (EOG) channels (right outer canthus–left ear (ROC-A1), left outer canthus–left ear (LOC-A1), and one or two electromyograms (EMG) changes (mental and/or mylohyoides). Neurological examination was performed again immediately before PSG.

PSG recordings were analyzed retrospectively by an expert electroencephalographer (F.P.) blind to the patients’ clinical data and outcomes. According to the type of sleep pattern found, patients were classified into 5 groups, in increasing order of sleep organization complexity:

1. Monophasic: continuous low voltage theta–delta activity (Fig. 1A);
2. Cyclic alternating pattern (CAP): absence of sleep elements in the presence of recurrent abnormal spontaneous arousal activity. The term CAP was used here because the cyclic alternation between two types of activity resembles that of the CAP, described by Terzano et al., (1985). In the present study, the term indicates a pattern where epochs characterized by low-voltage theta activity alternate with epochs with bursts of high-voltage monomorphic delta waves (Fig. 1B, C);
3. Rudimentary sleep: presence of rudimentary NREM sleep elements (K-complexes and/or spindles, Fig. 1D);
4. Non-rapid eye movement (NREM): presence of well-structured elements of NREM sleep (Fig. 1E);
5. Rapid eye movements (REM): presence of REM sleep elements (REMs and saw-tooth waves) alternating with NREM sleep (Fig. 1F).

Since different patterns could be present at different epochs of the same recording, the most organized pattern found in each recording was the one considered for the classification of each individual patient.

Although a few EEG arousals (characterized by delta bursts associated with increased muscular tone) could be found in patients from groups 4 and 5, a clear-cut wake pattern, with well-structured alpha activity, was never found.

Clinical outcome was assessed by evaluating the patient’s degree of autonomy with the Glasgow Outcome Scale (GOS) (Jennett and Bond, 1977) after an interval of at least 1 year (range 12–34 months). Patients were assigned to one of the 5 groups: (1) good recovery; (2) moderate disability; (3) severe disability impeding autonomous life; (4) persistent vegetative state; and (5) death.

Statistical analysis was performed correlating clinical outcome (GOS) with age, with GCS at the admission and at the subacute stage, with severity of central nervous system (CNS) damage according to neuroradiological find-

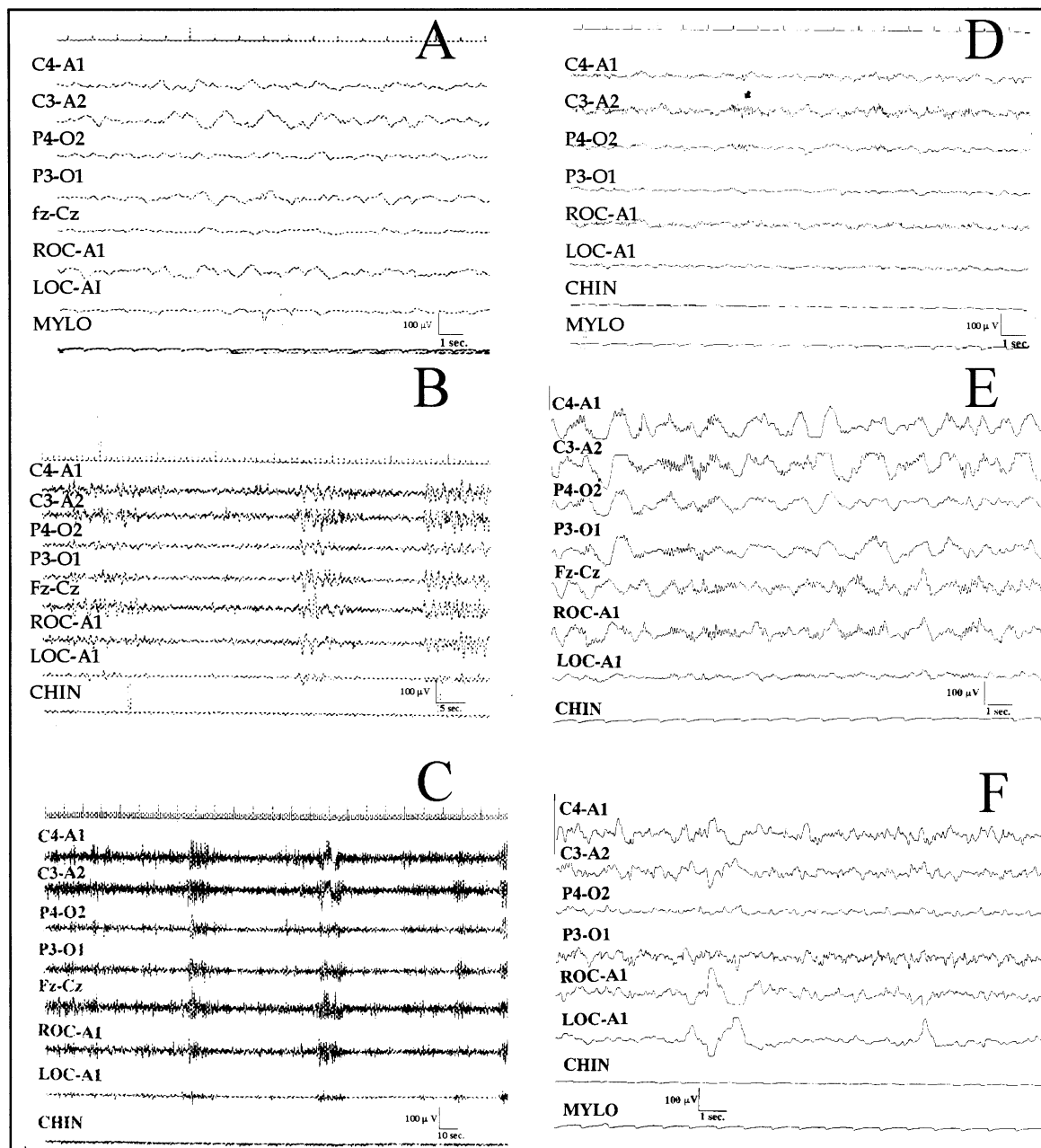


Fig. 1. Epochs of PSG recordings. (A) Epoch, 20 s, of monophasic pattern, characterized by monotonous, non-reactive, low-voltage theta-delta activity, without sleep elements during the entire 24 h recording interval; (B) 80 s epoch of CAP; (C) same patient as in (B), compressed epoch of 240 s, showing the cyclicity of the alternation between high-voltage monomorphic delta activity and low-voltage theta activity; (D) rudimentary sleep pattern, with the arrow showing an asymmetric and rudimentary sleep spindle; (E) NREM pattern, with well-structured sleep spindles and K-complexes; and (F) REM pattern (note the rapid eye movements registered at the EOG derivations).

ings and with polysomnographic pattern by means of Mantel–Haenszel test for linear association. The accuracy of each of these diagnostic tools in predicting the outcome was compared by plotting the respective ROC curves, which combine the sensitivity and specificity achieved by the diagnostic test at several cutoff points (Fletcher et al., 1996). In addition, a multiple regression analysis with clinical outcome as the dependent variable was also performed in order to identify the factors likely to predict outcome.

### 3. Results

Twenty-four patients (19 males, 5 females; mean age  $41 \pm 17.09$ , range 16–74) were studied. Clinical features of patients are shown in Table 1.

#### 3.1. Clinical outcome

(a) Death: 4 patients (16.7%).

Table 1

Clinical features of the studied patients, including GCS and GOS scores<sup>a</sup>

| Patient | Age (years) | Sex | Head Injury                                                                                                          | GCS at admission | GCS at the subacute phase | Sleep pattern on PSG | Follow-up (months)  | GOS |
|---------|-------------|-----|----------------------------------------------------------------------------------------------------------------------|------------------|---------------------------|----------------------|---------------------|-----|
| 1       | 47          | m   | L F–P skull fracture, uncal herniation, subdural hematoma, IVH                                                       | 5                | 10                        | 2                    | Dead after 2 months | 1   |
| 2       | 59          | m   | L P skull fracture, L T–P hematoma, subfalcial herniation                                                            | 6                | 7                         | 2                    | Dead after 2 months | 1   |
| 3       | 25          | f   | R T epidural hematoma, pneumoencephalus                                                                              | 4                | 9                         | 4                    | 34                  | 5   |
| 4       | 20          | m   | Hemorrhage in the R lenticular nucleus                                                                               | 8                | 12                        | 5                    | 34                  | 5   |
| 5       | 30          | f   | ssL P subdural hematoma                                                                                              | 7                | 7                         | 3                    | 31                  | 3   |
| 6       | 50          | m   | R P subdural hematoma, subfalcial herniation                                                                         | 3                | 15                        | 5                    | 31                  | 4   |
| 7       | 52          | m   | L T–P hematoma, monoventricular hydrocephalus, subfalcial herniation                                                 | 8                | 15                        | 4                    | 30                  | 4   |
| 8       | 19          | m   | F skull fracture, R F hematoma with invasion of basal ganglia, subfalcial herniation                                 | 3                | 5                         | 3                    | 26                  | 3   |
| 9       | 47          | f   | R T hemorrhage, multiple frontal hemorrhagic areas                                                                   | 7                | 6                         | 4                    | 24                  | 5   |
| 10      | 44          | m   | L F–P subdural hematoma                                                                                              | 8                | 12                        | 5                    | 23                  | 4   |
| 11      | 72          | m   | Intraventricular hemorrhage                                                                                          | 8                | 10                        | 2                    | 23                  | 3   |
| 12      | 58          | f   | Subarachnoid and intraventricular hemorrhage, brainstem ischemic lesion                                              | 4                | 4                         | 2                    | Dead after 3 months | 1   |
| 13      | 16          | m   | R T–P skull fracture, R F–P contusion, R carotid-cavernous fistula                                                   | 6                | 11                        | 5                    | 20                  | 3   |
| 14      | 45          | m   | R temporo-mastoid skull fracture, L subdural hematoma                                                                | 6                | 10                        | 4                    | 18                  | 4   |
| 15      | 28          | m   | Multiple ischemic lesions of subcortical white matter                                                                | 7                | 8                         | 5                    | 23                  | 5   |
| 16      | 46          | f   | Subdural hematoma                                                                                                    | 9                | 9                         | 3                    | 17                  | 3   |
| 17      | 38          | m   | L F–T Intraaxial hematoma                                                                                            | 3                | 6                         | 4                    | 18                  | 3   |
| 18      | 18          | m   | R T fracture, multiple hemorrhagic areas in cortico-medullar junction and in peri-mesencephalic/cerebellar cisternae | 6                | 7                         | 4                    | 16                  | 5   |
| 19      | 74          | m   | L T intraaxial hemorrhage                                                                                            | 3                | 3                         | 1                    | Dead after 3 months | 1   |
| 20      | 27          | m   | Cerebral swelling                                                                                                    | 4                | 11                        | 4                    | 15                  | 5   |
| 21      | 46          | m   | R P–O fracture; R P epidural hematoma                                                                                | 6                | 6                         | 2                    | 14                  | 3   |
| 22      | 26          | m   | R T fracture, R F epidural hematoma                                                                                  | 7                | 15                        | 5                    | 13                  | 5   |
| 23      | 41          | m   | R subdural hematoma, subfalcial hernia                                                                               | 4                | 10                        | 4                    | 13                  | 4   |
| 24      | 48          | m   | L T–P epidural hematoma; R T contusive focus                                                                         | 6                | 8                         | 2                    | 12                  | 2   |

<sup>a</sup> Abbreviation: m, male; f, female; F, frontal; T, temporal; P, parietal; O, occipital; L, left; R, right; IVH, intraventricular hemorrhage; and PSG, polysomnography.

Four patients eventually died after infectious complications. In particular, two of them (patients #1 and #2, Table 1) died after having come out of coma, during post-intensive admission in the neurosurgery department. The other two (patients #12 and #19, Table 1) died while still comatose in the ICU.

(b) Persistent vegetative state: one patient (4.2%).

(c) Severe disability: 7 patients (29.2%).

Five of the 7 patients were able to live a somewhat autonomous life, that is with some limitations and the need for assistance in some of their daily living activities, whereas two of them survived totally unable to lead any form of autonomous life.

(d) Moderate disability: 5 patients (20.8%).

Five patients regained autonomy, despite some minor neurological deficits.

(e) Good recovery: 7 patients (29.2%).

Seven patients recovered completely, with no disability.

### 3.2. Neuroimaging findings

According to the signs found in their CT scan, patients were divided into 4 groups, in decreasing order of severity: (a) patients with direct or indirect brainstem involvement (3 patients, 12.5%); (b) patients with signs of cerebral swelling, except those assigned to group (a) (5 patients, 20.8%); (c) patients with parenchymal lesions, as cortical contusions, intraparenchymal hematomas, shear injuries, etc., with the exclusion of those assigned to groups (a) and (b) (8 patients, 33.3%); and (d) patients with signs of extra axial lesions, as epidural and subdural hematomas, except those assigned to group (a), (b) or (c) (8 patients, 33.3%).

### 3.3. PSG pattern

#### 3.3.1. Group 1: monophasic pattern

Only one patient (4.2%) showed continuous theta–delta activity, with persistent low voltage, without spontaneous arousals throughout the entire recording. This patient, with a left temporal hemorrhage, died 3 months after injury due to septic complications, while still comatose in the ICU.

#### 3.3.2. Group 2: CAP

Six patients (25%) showed an oscillatory pattern, changing from high voltage theta–delta activity to low voltage mixed frequency activity. Three of them died; another one was in a persistent vegetative state, whereas the other two remained severely disabled.

#### 3.3.3. Group 3: rudimentary sleep

Three patients (12.5%) presented rudimentary NREM sleep elements, such as abnormal or atypical morphology, frequency and scalp distribution of sleep spindles and/or K-

complexes. These patients had an intermediate outcome (severe disability).

#### 3.3.4. Group 4: NREM

Eight patients (33.3%) showed NREM sleep elements (i.e. K-complexes and sleep spindles normally formed). Seven of them (87.5%) had a good recovery; the last one remained disabled.

#### 3.3.5. Group 5: REM

REM sleep elements alternating with NREM sleep elements were detected in 6 patients (25%), of whom 5 (83.3%) showed good recovery. In particular, two of the 5 patients presented a high percentage of REM sleep, and the other 3 presented some epochs characterized by a mixture of REM sleep and NREM sleep elements. The sixth patient with the EEG pattern (16.7%) had an intermediate outcome (severe disability).

### 3.4. Correlations with clinical outcome

Clinical outcome, evaluated with the GOS, was significantly correlated with the PSG pattern ( $P < 0.001$ , Mantel–Haenszel test for linear correlation) (Table 2). In fact, better outcomes were related to better PSG patterns characterized by the presence of typical and well-structured sleep elements (REM and/or NREM). In addition, outcome showed a significant inverse correlation with age ( $P < 0.005$ , Mantel–Haenszel test for linear correlation) and a direct correlation with GCS at the end of the acute period ( $P < 0.05$ , Mantel–Haenszel test for linear correlation).

Neither GCS at admission, nor severity of the anatomical lesion, as shown by CT scans, showed linear correlation to the outcome (GOS).

Multiple regression analysis was done with clinical outcome as the dependent variable, tested for possible correlation with the variables to which it was significantly correlated in univariate analysis (i.e. age, GCS at the subacute stage and PSG pattern). The results of this analysis showed that only PSG pattern was independently associated with GOS ( $P = 0.01$ , adjusted odds ratio (OR) = 10.78).

The finding of organized sleep patterns predicted a good outcome (full recovery or only mild disability) with a positive predictive value of 86%, sensitivity of 100%, and specificity of 83%. The accuracy of PSG patterns to predict a favorable outcome was higher than that of GCS, age and neuroimaging, as shown by comparison of ROC curves (Fig. 2).

The same calculations were repeated after having excluded from the group the two patients who died of non-neurological causes after having recovered from coma (patients #1 and #2, Table 1), as well as the 3 patients with a GCS at the subacute stage equal or higher than 13 (patients 6, 7, and 22, Table 1). The selection of cases was aimed at excluding a possible confounding variable (death by infec-

Table 2  
Relationship between final outcome and PSG pattern ( $n = 24$ )<sup>a</sup>

| Outcome                     | REM          | NREM         | Rudimentary NREM | CAP          | Monophasic  |
|-----------------------------|--------------|--------------|------------------|--------------|-------------|
| Full recovery               | 3<br>(50%)   | 4<br>(50%)   | –                | –            | –           |
| Mild disability             | 2<br>(33.3%) | 3<br>(37.5%) | –                | –            | –           |
| Severe disability           | 1<br>(16.7%) | 1<br>(12.5%) | 3<br>(100%)      | 2<br>(33.3%) | –           |
| Persistent Vegetative state | –            | –            | –<br>(16.7%)     | 1            | –           |
| Death                       | –            | –            | –                | 3<br>(50%)   | 1<br>(100%) |

<sup>a</sup>  $\chi^2 = 15.18$ , one degree of freedom,  $P = 0.0001$  (Mantel–Haenszel test for linear association).

tious complications, unrelated to the neurological problem), and at eliminating those patients whose prognosis could be considered favorable on the sole basis of the clinical examination. The selected group included 19 patients with an undetermined prognosis on the basis of clinical examination, which constitutes the real problem in daily practice. Results were similar to those found in the unselected group, with sleep pattern at PSG showing higher accuracy to predict outcome than GCS, age or neuroimaging. Again,

the finding of sleep patterns (NREM or REM) predicted a good outcome with a positive predictive value of 82%, sensitivity of 100%, and specificity of 80%.

#### 4. Discussion

In the past few years, some studies considered sleep pattern as a prognostic criterion in post-traumatic coma (Bergamasco et al., 1968a,b; Bricolo and Turella, 1973; Bricolo et al., 1979). An important limitation of previous investigations, however, lies in the fact that clinical outcome was considered dichotomically, on the basis of survival or exitus, without considering, in case the patients survived, their final level of functional independence. Our study demonstrates that the PSG pattern is a reliable prognostic marker in head injury coma, with a particularly high predictive value for survival. In fact, none of the patients presenting sleep elements (not only well-structured ones, but also rudimentary ones) died, whereas 4 out of 7 (57%) of the patients who had only CAP or monophasic patterns died. However, more important than that is the fact that the predictive value of PSG monitoring is also high for functional recovery. In fact, 86% of the patients who presented well-structured NREM and/or REM sleep elements showed a good recovery, with no sequelae or with only mild neurological deficits. On the contrary, the 3 patients who survived in the group of 6 patients showing only CAP during the entire monitoring presented severe disabilities or persistent vegetative state.

This finding concerning CAP contrasts with the study of Evans and Bartlett (1995), who attributed a good prognosis to this pattern. A possible explanation for this difference could be that some sleep elements could have been missed in their study because of the short duration of their recordings (30–60 min), which prevents the identification of circadian oscillations. However, it must be taken into consideration the fact that two of the 3 patients with CAP died of infectious complications after having come out of

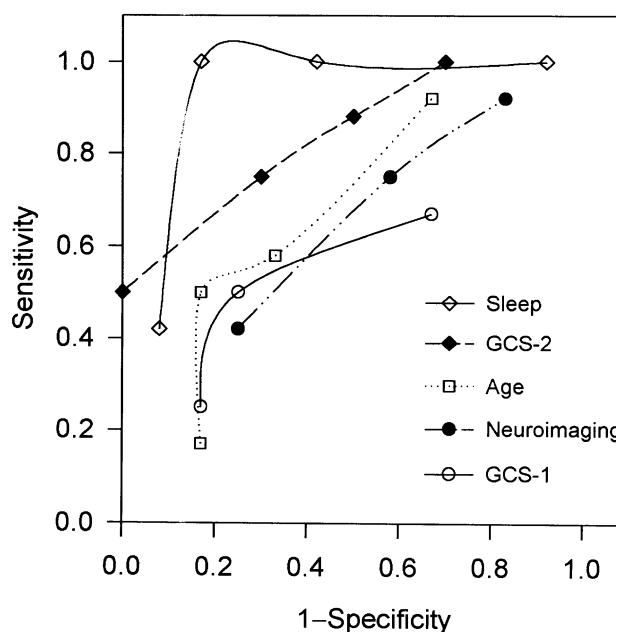


Fig. 2. Receiver operator characteristics (ROC) curves showing accuracy for prediction of good outcome (good recovery or only moderate disability) after post-traumatic coma according to the findings of 24 h PSG monitoring (sleep), GCS scores at admission (GCS1) or at the subacute phase (GCS2), age, and neuroimaging findings. The ordinate axis shows the sensitivity of the tests, ranging from 0 to 1.0 (0–100%), while the abscissa shows the percentage of false positive results (1-specificity). Tests with good discriminatory power should concentrate on the left upper corner of the graph (high sensitivity combined with high specificity) (Fletcher et al., 1996).

coma. As we discuss later, it might be suggested that the infectious complications could be a consequence of immunological impairment due to the neurophysiopathological problem. Nevertheless, considering that this is solely an unproven hypothesis, we recalculated the predictive value of the EEG after having excluded these patients who died of presumed unrelated causes, and came to the same result: the presence of CAP all along the 24 h of recording at the subacute stage of post-traumatic coma still predicted a bad prognosis (death or severe disabilities).

The presence of rudimentary NREM sleep elements seemed to be a good prognostic factor only for survival but not for functional recovery, since all the patients with this pattern survived, but with severe disability. The existence of qualitatively abnormal K-complexes or spindles could be an index of a severe alteration of CNS.

Although our study sample does not allow to establish the precise value of each particular sleep pattern to predict the outcome in post-traumatic coma, our data strongly indicate that the presence of well-structured patterns of NREM and/or REM sleep identifies patients with higher probability of favorable outcomes (good recovery or moderate disability). The same predictive value was observed when the analysis was repeated excluding the patients whose prognosis could be considered favorable on the sole basis of the clinical examination (i.e. those with GCS at the subacute stage equal or higher than 13), which suggests that PSG could be useful in the clinical situation where prognosis remains undetermined on the basis of the clinical examination.

This association between sleep pattern and clinical outcome persisted after multivariate analysis correction for age and GCS, proving that it was not generated by some confounding bias.

GCS is currently considered as the gold standard for outcome prediction after head injury (Jennett et al., 1976). In the present study, GCS at admission does not seem to be an accurate predictor of outcome, at least regarding the degree of disability. It may have some prognostic value for survival, based on the fact that no patient with GCS scores higher than 7 at the acute phase died, but GCS at the acute phase did not show a significant correlation with outcome scores. When assigned after the end of the acute phase, GCS scores did show a significant linear correlation with GOS. This positive correlation seems to be favored by the fact that extreme values are associated with clear-cut outcome results. Thus, all patients with GCS after the end of the acute phase lower than 5 died, while all patients with GCS at that moment higher than 11 had a favorable prognosis, with good recovery or moderate disability. Patients with intermediate values of GCS (GCS ranging from 5 to 11,  $n = 17$  out of 24), however, showed different, scattered outcome results at follow-up. Multivariate analysis failed to show an independent association between outcome and GCS at the subacute phase.

The same pattern of correlation was verified regarding age. The prognostic relevance of age in severe head-injured

patients has been well established in previous studies (Jennett et al., 1976; Jennett and Teasdale, 1981; Evans and Bartlett, 1995). Our study confirms that younger age is usually associated with better outcome, but age per se cannot be considered as an independent prognostic marker, as shown by multivariate analysis.

Finally, according to the present data, the severity of neuroradiological findings measured by cerebral CT scans failed to predict clinical outcome. It is possible that this lack of correlation between the anatomical damage and the outcome was due to the limited sensitivity of CT as compared to cerebral magnetic resonance imaging (MRI) (Kampfl et al., 1998).

In summary, sleep pattern organization revealed a better predictor of outcome in post-traumatic coma when compared to other classical parameters such as GCS or neuroimaging. Indeed, PSG studies evaluate the functional integrity of the CNS more so than neuroimaging studies or GCS. It would stand to reason that preserved cycling of NREM and REM sleep would imply better functional integrity of the CNS, and therefore, would and should be a favorable prognostic finding.

In all the patients who died in our case series, the fatal outcome was due to infectious complications. In two of them, the fatal outcome took place after post-intensive admission to the department of neurosurgery, after having come out of coma and having regained autonomous ventilation. The two other patients who died never came out of coma. In the 4 patients who died, serial PSG recordings continued to document the absence of sleep elements, showing only CAP in 3 of them, including the two patients who were out of coma, and monophasic pattern in the patient whose lesions involved even the brainstem. It is reasonable to wonder whether the severe sleep disruption did not play a specific role in – or at least, if it does not represent an important sign of – decreasing immunitary responses in these patients (despite temporarily getting out of coma for two of them). Indeed, many clinical and experimental studies in the literature suggest that there is a relationship between sleep and the proper functioning of host defense systems (Benca and Quintans, 1997; Ader et al., 1995; Dinges et al., 1995; Rechtschaffen and Bergmann, 1995).

In conclusion, the pattern of sleep organization, as determined with the aid of prolonged PSG, showed the highest predictive value among the considered variables. It is most likely that the functional response of the CNS to injury is influenced by age and by the severity of the anatomical damage, but it is certainly a complex, multi-factorial process. The currently available tools for the evaluation and prediction of outcome may reflect, in a more or less reliable way, what this functional response is likely to be. The sleep pattern detected by 24 h PSG monitoring after the end of the acute phase seems to be, as demonstrated by the present data, a very reliable prognostic marker in head injury coma, with a much higher predictive value both for survival and for functional rehabilitation than GCS alone.

## References

- Ader R, Cohen N, Felten D. Psychoneuroimmunology: interactions between the nervous system and the immune system. *Lancet* 1995;345:99–103.
- Benca RM, Quintans J. Sleep and host defenses: a review. *Sleep* 1997;20(11):1027–1037.
- Benzer A, Mitterschiffthaler G, Marosi M, Luef G, Puhlinger F, De La Renotiere K, Lehner H, Schmuthzhard E. Prediction of non-survival after trauma: Innsbruck Coma Scale. *Lancet* 1991;338(8773):977–978.
- Bergamasco B, Bergamini L, Doriguzzi T, Fabiani T. EEG sleep patterns as a prognostic criterion in post-traumatic coma. *Electroenceph clin Neurophysiol* 1968a;24:374–377.
- Bergamasco B, Bergamini L, Doriguzzi T. Clinical value of the sleep electroencephalographic patterns in post-traumatic coma. *Acta Neurol Scand* 1968b;44:495–511.
- Bricolo A, Turazzi S, Faccioli F. Combined clinical and EEG examinations for assessment of severity of acute head injuries. *Acta Neurochir* 1979;Suppl 28:35–39.
- Bricolo A, Turella G. Electroencephalographic patterns of acute post-traumatic coma: diagnostic and prognostic value. *J Neurosurg Sci* 1973;17:278–285.
- Chatrian GE, White LE, Daly D. Electroencephalographic patterns resembling those of sleep in certain comatose states after injuries to the head. *Electroenceph clin Neurophysiol* 1963;15:272–280.
- Dinges DF, Douglas SD, Hamarman S, Zaugg L, Kapoor S. Sleep deprivation and human immune function. *Adv Neuroimmunol* 1995;5:97–110.
- Evans BM, Bartlett JR. Prediction of outcome in severe head injury based on recognition of sleep related activity in the polysomnographic electroencephalogram. *J Neurol Neurosurg Psychiatry* 1995;59:17–25.
- Fischgold H, Mathis P. Obnubilations, comas et stupeurs. Études électro-encephalographiques. *Electroenceph clin Neurophysiol* 1959; (Suppl 11), p. 125.
- Fletcher RH, Fletcher SW, Wagner EH, editors. *Clinical epidemiology: the essentials* 3rd ed. Baltimore, MA: Williams and Wilkins, 1996.
- Giubilei F, Formisano R, Fiorini M, Vitale A, Faroni J, Toni D, Santilli V. Sleep abnormalities in traumatic apallic syndrome. *J Neurol Neurosurg Psychiatry* 1995;58:484–486.
- Harada M, Minami R, Hattori E, Nakamura K, Kabashima K. Sleep in brain damaged patients: an all night study of 105 cases. *Kumamoto Med J* 1976;29:110–127.
- Jennett B, Bond M. Assessment of outcome after severe brain damage: a practical scale. *Lancet* 1977;1:480–484.
- Jennett B, Teasdale G, Brackman R. Predicting outcome in individual patients after severe head injury. *Lancet* 1976;1:1031–1034.
- Jennett B, Teasdale G. Prognosis after severe head injury. In: Jennett B, Graham D, editors. *Management of head injuries*, Philadelphia, PA: Davis, 1981. pp. 317–332.
- Kampfl A, Schmutzhard E, Franz G, Pfausler B, Haring HP, Ulmer H, Felber A, Golaszewski S, Aichner F. Prediction of recovery from post-traumatic vegetative state with cerebral magnetic-resonance imaging. *Lancet* 1998;351(9118):1763–1767.
- Rechtschaffen A, Bergmann BM. Sleep deprivation in the rat by the disk-over-water method. *Behav Brain Res* 1995;69:55–63.
- Ron S, Algom D, Hary D, Cohen M. Time-related changes in the distribution of sleep stages in brain injured patients. *Electroenceph clin Neurophysiol* 1980;48:432–441.
- Rumpl E, Prugger M, Bauer G, Gerstenbrand F, Hackl JM, Pallua A. Incidence and prognostic value of spindles in posttraumatic coma. *Electroenceph clin Neurophysiol* 1983;56:420–429.
- Terzano MG, Mancina D, Salati MR, Costani G, Decembrino A, Parrino L. The cyclic alternating pattern as a physiologic component of NREM sleep. *Sleep* 1985;8:137–145.